



Name: _____

Date: _____

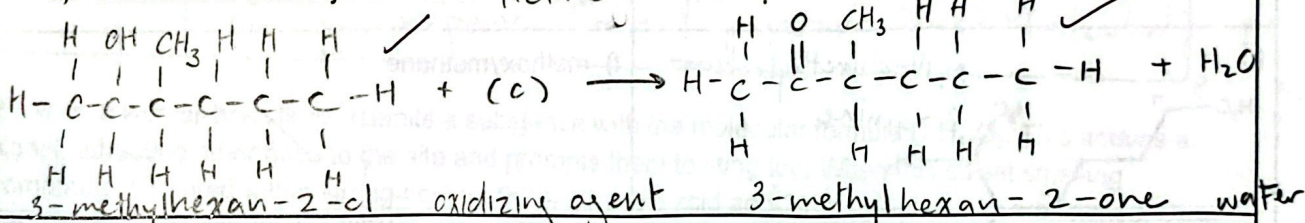
Chapter 1 Quiz

Knowledge (10 marks) – Multiple Choice on D2L

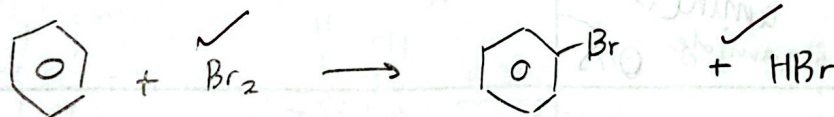
Thinking (10 marks)

1. Write an equation, using structural formula or carbon skeleton, for each of the following reactions and name the reactants and products: (8 marks)

a) Oxidation of 3-methylhexan-2-ol *ketone*

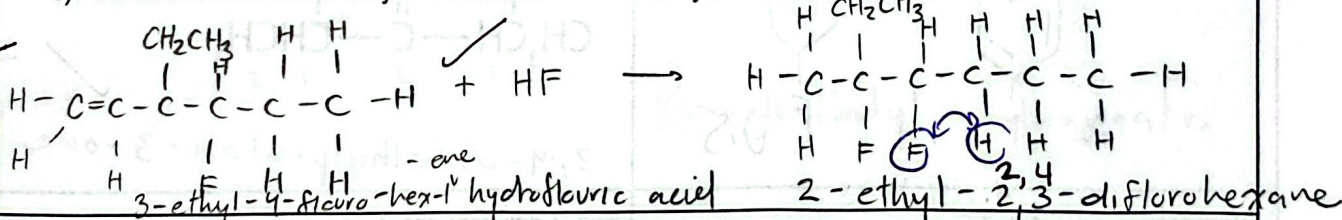


b) Reaction between benzene and bromine gas $\text{C}_6\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_6\text{H}_5\text{Br} + \text{HBr}$

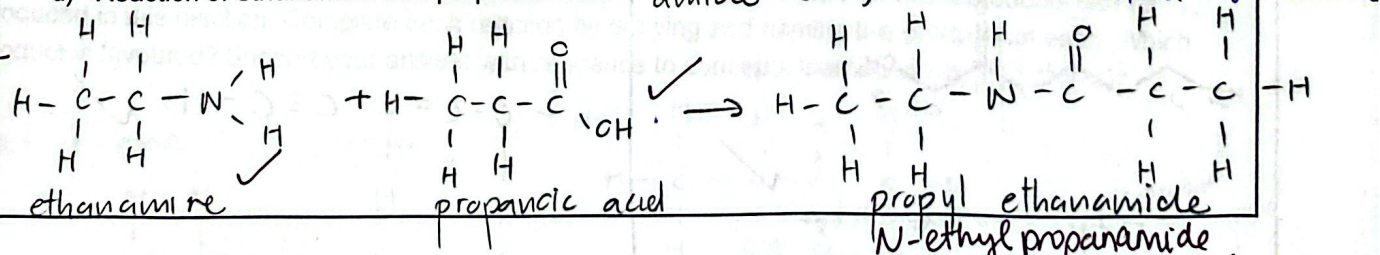


benzene bromine gas 1-bromobenzene hydrogen bromide

c) Reaction between 3-ethyl-4-fluorohex-1-ene and hydrofluoric acid



d) Reaction of ethanamine and propanoic acid



2. List the following 3 organic compounds in order of increasing boiling point. Explain in terms of intermolecular forces: dimethylether, ethane, ethanol (2 marks)

Lowest boiling point	ethane	dimethylether	ethanol	Highest boiling point
Explanation	ethanol has -OH group $\rightarrow$ H-bonding (1 BP) dimethylether has O-C $\rightarrow$ dipole dipole bonding (mid BP) ethane has C-H $\rightarrow$ LDF (1 BP)			

6.5

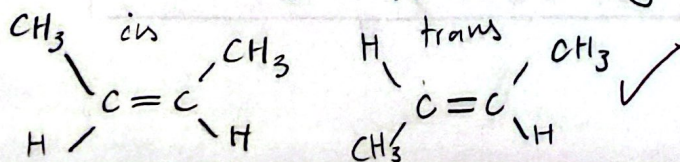
Communication (10 marks)

1. Complete the table below by naming the structural diagram for each of the organic compounds or drawing a structural diagram or carbon skeleton for each organic compound. (8 marks)

a) ethyl propanoate <i>ester</i>	e) <i>aldehyde</i> 2-methylpropanal 2-methylpropanal <i>propanal</i> 0.5
b) <i>N,N-dimethyloctan-3-amine</i> <i>N,N-dimethyloctan-3-amine</i> 0.5	f) methoxymethane
c) 2,4-difluorophenol 0.5	g) <i>ketone</i> 2,4-dimethylpentan-3-one
d) <i>ester</i> <i>butyl propanoate</i> ✓	h) 1,2-difluoropent-2-ene-4-yne

2. Briefly explain the difference between a cis isomer and a trans isomer. Use a diagram to help. (2 marks)

2 cis means the alkyl/hydrogen groups on an alkene are on the same side (horizontally) while trans means they are diagonally opposite (they have same formula, diff structure)

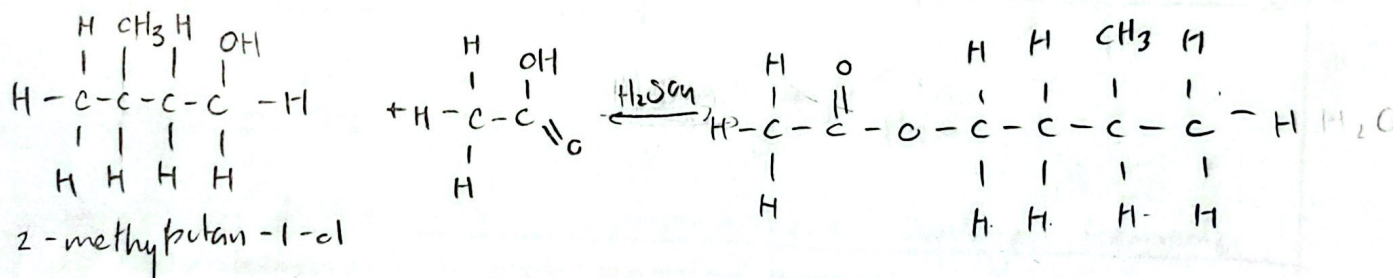


Application (9 marks)

1. Draw two compounds from 2 different organic families but both share the chemical formula $C_5H_{12}O$. Provide IUPAC name for each compound you draw. (4 marks)

Structure	$ \begin{array}{ccccccc} & H & H & H & & H & H \\ & & & & & & \\ H & -C & -C & -C & -O & -C & -C & -H \\ & & & & & & \\ & H & H & H & & H & H \end{array} $	$ \begin{array}{ccccccc} & H & OH & H & H & H \\ & & & & & \\ H & -C & -C & -C & -C & -C & -H \\ & & & & & \\ & H & H & H & H & H \end{array} $
Name	ethoxy propane (ether)	pentan-2-ol (alcohol)

2. When a worker-bee stings, it emits a substance with the molecular formula $C_7H_{14}O_2$. This acts as a signal, attracting other bees to the site and prompts them to sting too. When this sweet smelling compound is treated with a strong acid (H_2SO_4), ethanoic acid and 2-methylbutan-1-ol are formed. Draw the structure and provide the IUPAC name of the compound that is emitted by the bee. (2 marks)



2-methylbutyl ethanoate

3. The reaction of but-1-ene with water produces an alcohol; however, two different products can be produced in this reaction. Complete each reaction by drawing and naming the products of each. Which product is favoured? Support your answer with reference to concepts learned in class. (3 marks)

